AE-RAGX: Combining Autoencoders with Retrieval-Augmented Generation for Explainable Anomaly Detection using LLMs

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***Abstract*—Detecting anomalies in high-dimensional or unstructured data remains a challenge, as many traditional approaches such as Autoencoders (AEs), Variational Autoencoders (VAEs), and One-Class SVMs rely heavily on reconstruction errors or distance measures. These techniques often overlook contextual information, struggle with subtle deviations, and provide limited interpretability. To address these gaps, we propose AE-RAGX, a hybrid architecture that combines Autoencoders with Retrieval-Augmented Generation (RAG) using Large Language Models (LLMs). By transforming structured data into descriptive text and incorporating rule-based knowledge, the model not only improves anomaly detection but also produces transparent, human-readable explanations. Evaluation on a credit card fraud dataset shows that AE-RAGX enhances recall and provides interpretable outputs, making it well suited for high-stakes applications such as fraud detection.**

***Index Terms*—Anomaly detection, large language model (LLM), retrieval-augmented generation (RAG).**

I. Introduction

Anomaly detection identifies data points that deviate from normal patterns, with applications in fraud detection, cancer diagnosis, and cybersecurity monitoring [1]. A major challenge is the scarcity of labeled anomaly data, which limits supervised methods and motivates unsupervised or semi-supervised approaches trained on normal data. Another limitation is the lack of explainability; in critical domains like fraud detection, understanding why an anomaly is flagged is as important as detecting it, since missed anomalies can cause severe losses [1].

Autoencoders (AEs) are neural networks that reconstruct input data and flag anomalies when reconstruction error exceeds a threshold. Despite their effectiveness, they face several drawbacks: sensitivity to small input changes that allow adversarial evasion, instability during training, and better performance on simple than complex data, which can cause missed anomalies [2]. Moreover, relying on a single overall error may mask important local errors, making threshold selection difficult [3].

Variational Autoencoders (VAEs) adopt a probabilistic approach by modeling reconstruction probability but can fail when anomalies resemble normal patterns [5]. One-Class SVMs (OC-SVMs) learn a boundary around normal data but struggle with high-dimensional datasets (curse of dimensionality) and require complex hyperparameter tuning [6]. Ensemble Autoencoders (EoAEs) enhance robustness by combining multiple AEs, yet face challenges such as maintaining model diversity, higher computational cost, and determining the optimal number of models.

More broadly, traditional machine learning techniques struggle with the high non-linearity and evolving nature of modern datasets. These methods often require intensive manual feature engineering, which may not generalize well [1].

A. Our Contribution

In response to these challenges, our project introduces AE-RAGX, a novel hybrid architecture that addresses three critical limitations of traditional methods: (1) lack of explainability, (2) inability to capture contextual information and long-range dependencies, and (3) high false negative rates in critical applications like fraud detection. To highlight our key contributions, we summarize them as follows:

a.) Pioneering LLM Application: We introduce one of the first applications of LLMs for anomaly detection, shifting the focus from numerical reconstruction to semantic understanding of data irregularities.

b.) Novel LLM-Based Architecture: We propose an architecture that embeds LLMs into the anomaly detection pipeline. By interpreting structured input data through natural language, the model is better equipped to detect subtle deviations.

c.) Explainability: A major advantage of our approach is its explainability. The LLM can articulate why an input is classified as anomalous using human-readable descriptions. This enhances interpretability and trust, which is essential in sensitive domains like healthcare or finance.

d.) Opening New Research Directions: Our work opens new paths for using LLMs in data analysis, creating anomaly detection systems that adapt to complex and changing environments. Like an art historian spotting forgeries by understanding style and story, not just surface flaws, our model finds anomalies by combining data patterns with deeper contextual, logical, and sequential understanding.

This paper introduces AE-RAGX, a new anomaly detection method using Large Language Models. It identifies irregularities through understanding context and meaning, improving accuracy, clarity, and adaptability compared to traditional

TABLE I
FEATURE COMPARISON: AE-RAGX VS. STATE-OF-THE-ART MODELS

Feature	Autoencoder	VAE	OC-SVM	AE-RAGX
Detection Mechanism	Reconstruction error from compressed representation	Probabilistic reconstruction likelihood	Boundary separation in feature space	AE reconstruction + LLM contextual reasoning
Context Awareness	Limited	Limited	None	High – uses RAG and rules for context
Explainability	None	None	None	Human-readable, rule-based justifications
Scalability	Good	Good	Poor in high dimensions	LLM inference adds latency

statistical or reconstruction-based approaches.

The paper is structured as follows: Section II reviews related work, Section III presents the AE-RAGX architecture, Section IV explains the implementation, Section V reports experimental results, and Section VI provides conclusions and future directions.

II. RELATED WORKS

Anomaly detection is the process of finding observations that differ greatly from the rest, often caused by a different mechanism. It is vital in areas like fraud detection, cancer diagnosis, network security, and complex system monitoring [1]. A major challenge is the lack of labeled anomaly data, which limits supervised methods and leads to the use of unsupervised and semi-supervised approaches trained mainly on normal data.

A. Anomaly Detection Mechanisms and Limitations

Numerous machine learning and deep learning models have been developed for anomaly detection, each with distinct mechanisms and inherent limitations.

Autoencoders: Autoencoders (AEs) are neural networks that reconstruct input data, using reconstruction error as the anomaly score [3]. Trained on normal data, they should give low errors for normal inputs and higher for anomalies. However, they are vulnerable to adversarial attacks that can reduce anomaly errors, and exhibit a complexity bias, often reconstructing simpler anomalies as well as normal data [2]. Summing errors across features can also hide local anomalies, so one improvement is to use reconstruction error as a vector with feature-specific thresholds.

Variational Autoencoders: Variational Autoencoders (VAEs) are probabilistic models that use reconstruction probability as an anomaly score, providing a principled measure over simple reconstruction error by accounting for data variability [8]. However, they may miss anomalies with simple structures derived from normal data, e.g., failing to detect MNIST digits '1', '7', and '9' [9]. Their main limitation lies in poor explainability and lack of contextual awareness. In fraud detection, where false negatives risk major financial losses, it is vital to understand why a transaction is flagged and ensure full detection coverage.

One-Class SVM: One-Class SVM is a kernel-based method that learns a boundary around normal data by separating it from the origin in a transformed feature space. While effective, it struggles with large, high-dimensional datasets and is sensitive to parameter choices like the *nu* value, which affects margin errors and support vectors [10].

Traditional machine learning methods like SVM, Naïve Bayes, and Random Forest have been used for intrusion detection but often struggle with the high non-linearity and fast-changing nature of modern attacks. They also require heavy manual feature engineering, which may not work well across different or evolving threats [11].

B. LLMs for Anomaly Detection

Our project introduces anomaly detection with Large Language Models (LLMs), a novel and underexplored approach. Unlike traditional methods that rely on numerical patterns or statistical boundaries, LLMs translate structured data into descriptive text, enabling contextual reasoning and human-readable explanations. This improves transparency and interpretability, especially in sensitive domains like healthcare and cybersecurity [12]. Table I compares traditional methods (Autoencoders, VAEs, OC-SVMs) with our AE-RAGX model. As an analogy, Autoencoders act like guards flagging items unlike known exhibits, OC-SVMs set invisible boundaries but falter in complex cases, Ensemble AEs use multiple guards with coordination issues, while LLMs work like expert analysts—spotting anomalies and explaining why they are suspicious.

III. THE PROPOSED AE-RAGX : AUTOENCODER WITH RETRIEVAL-AUGMENTED GENERATION AND EXPLAINABILITY ARCHITECTURE

While autoencoders detect anomalies via reconstruction errors, they lack contextual awareness, long-range dependency handling, and explainability. To address this, we propose AE-RAGX, which combines autoencoders with Retrieval-Augmented Generation (RAG)-based LLMs to add contextual reasoning and generate transparent outputs. The architecture has two stages: (1) *Autoencoder-Based Reconstruction Error Analysis*, which learns normal patterns and flags anomalies statistically, and (2) *Context-Aware Anomaly Detection using RAG-Based LLM*, which incorporates domain rules to enhance reasoning and explanations. As shown in Figure 1, these stages are further divided into sub-phases, with detailed implementations described in Section IV.

Phase 1 (Data Processing Phase): First, we obtain the transaction data and perform preprocessing to extract relevant features for the Autoencoder (AE). The AE then processes these features and generates a reconstruction error.

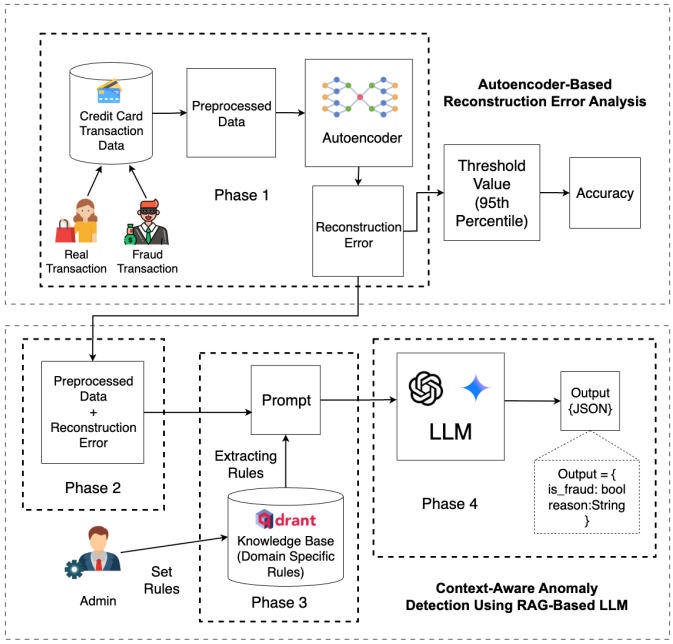

Fig. 1. AE-RAGX Architecture

Phase 2 (Text Generation Phase): Using both the reconstruction error from the AE and the preprocessed transaction data, we generate a natural language description of the transaction. This text representation makes the transaction data more suitable for LLM for transaction analysis.

Phase 3 (Knowledge Retrieval Phase): The system maintains a knowledge base containing predefined rules. They are stored in a vector database for efficient retrieval. When processing a transaction, relevant rules are retrieved from this vector store based on the transaction query similarity.

Phase 4 (Analysis and Decision Phase): Finally, the generated transaction text, along with a structured prompt, is fed into a Large Language Model (LLM). The LLM analyzes the transaction description against the retrieved rules from the vector store. This produce a final output that includes not only the decision but also the relevant reasoning and explanations for that decision.

IV. Implementation

This section details the practical realization of the AE-RAGX architecture described above.

A. Autoencoder-Based Reconstruction Error Analysis

1) Credit Card Transaction Dataset: For our experiment, we used a credit card transaction dataset containing 1.2 million records of both genuine and fraudulent transactions [13]. The dataset includes 24 features spanning numerical attributes such as transaction amount, geographical coordinates (latitude and longitude), and city population, as well as categorical attributes like merchant type, transaction type (e.g., grocery, entertainment), cardholder gender, state, and other relevant details.

2) Data Preprocessing, Feature Engineering and Feature Separation: Data preprocessing involved scaling and encoding: numerical features were normalized using standard and min-max scalers, while categorical features were one-hot encoded. Feature engineering added domain-inspired signals, such as transaction counts, average amounts over time windows (spending habits), and distance-based measures for unusual geographical activity. These rule-based features enhance contextual information for anomaly detection.

- Total transaction counts for the past 30, 7, and 1 days, and the past 1 hour
- Average transaction amount for the past 30, 7, and 1 days, and the past 1 hour
- Distance between the merchant address and the user address
- num of transactions in weekend/(num of transactions in weekday + num of transactions in weekend)

These features help the model better understand user behavior trends over time. One-hot encoded columns are separated from the test dataset, as LLMs understand raw categorical information better [14]. So, while the encoded numerical features are fed into the autoencoder, the raw categorical and context-rich data are passed to the LLM in the next stage.

3) Autoencoder Training: After we implement the Autoencoder, it is trained exclusively on pre-processed normal (non-fraudulent) transaction data. The purpose of training with non-fraudulent data is to generally capture the pattern in normal data and then detect the anomalies.

The autoencoder architecture includes:

- Input layer with 45 neurons
- A compressed latent representation with 2 neurons
- A decoder that reconstructs the original 45-dimensional input from the latent space

This architecture ensures that the model captures essential features while discarding noise.

4) Reconstruction Error Calculation: Once trained, the autoencoder is used to reconstruct data points. The reconstruction error is computed as the difference between the original and reconstructed values, typically using mean squared error. This reconstruction error is the core part behind the autoencoder, as it gives a large error for the data points deviated from the general pattern.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{1}$$

5) Threshold Determination: As a threshold is needed to establish a benchmark for flagging data points based on the autoencoder prediction, we decide on a threshold value. A comparison is done by selecting the 85th, 90th, and 95th percentiles of reconstruction errors observed in the testing data. Transactions with errors above this threshold are flagged as potential anomalies. Among the percentile values, the 95th percentile performed better, and we used it as our threshold value. The results for all three threshold values can be found in the Experimental Evaluation section.

6) Evaluation Metrics: The autoencoder output is evaluated using standard metrics such as accuracy, precision, recall, and f1-score. These metrics quantify the effectiveness of the autoencoder in isolating anomalies.

B. Context-Aware Anomaly Detection Using RAG-Based LLM

1) Rule-Based Knowledge Integration: In this stage we have brought back the missing context by defining 25 domain-specific rules that were curated through dataset analysis and background research on rule-based expert systems [15]. The referenced work [15] describes the development of a rule-based expert system for detecting credit card fraud during the authorization process. In that study, rules were created through expert consultation with bank managers, who identified predictive variables such as transaction amount, time of day, merchant type, and transaction frequency. These variables were then paired with defined thresholds to flag suspicious behavior, achieving over 89% classification accuracy and significant cost savings. Inspired by this approach, we adapted similar principles when designing our rules. As shown in Table II, some of the selected rules cover different aspects of transaction behaviour.

TABLE II
SELECTED RULES FOR TRANSACTION FRAUD DETECTION

Rule ID	Description
Rule 1	Flag if the transaction Amount is more than 10 times the average over the past 30 days. Reason: Unusual high-value spending. Action: Set risk to high. Category: Amount-Based.
Rule 4	Flag if transaction occurs between 2–6 AM and no prior activity in that time. Reason: Outside usual activity hours. Action: Set risk to high. Category: Time-Based.
Rule 11	Flag if transactions occur in cities is more than 100 miles apart within 2 hours. Reason: Implausible travel speed. Action: Set risk to high. Category: Location-Based.
Rule 20	Flag if transaction is in a category never used before (e.g., jewelry for a user who shops only groceries). Reason: Unusual category. Action: Set risk to high. Category: Category-Based.

We have derived rules based on 9 categories, such as amount-based, frequency-based, time-based, transaction location-based, merchant-based, transaction type-based, cardholder behaviour-based, cardholder detail-based, and reconstruction error-based. These rules will help to look into the missing context of the autoencoder.

2) RAG Pipeline: We implement the RAG pipeline by combining the rule vector database with a transformer-based language model. Rules act as contextual knowledge for the prompt, while the transaction data and its reconstruction error are included as part of the input.

Referring to the LLMs, we have used gpt-4o model by OpenAI and gemini-2.5-pro model by Google. Then, to get the prediction and explanation separately pydantic output parser has been used and get a structured output.

3) Final Evaluation: The hybrid system is evaluated using accuracy, precision, recall, and F1 score, enabling direct comparison with the standalone autoencoder and highlighting the benefits of added context for improved detection.

V. EXPERIMENTAL EVALUATION

To evaluate the performance of the AE-RAGX and to detect fraudulent transactions, it is compared with the autoencoder model. The evaluation metrics are Accuracy, Precision, Recall, and F1 score with confusion matrices. For testing, 50 transaction samples with the same amount of fraud and clean transactions were selected.

1) Performance Metrics: The AE-RAGX shows a significant improvement in recall over the AE model. In fraud detection, recall is particularly important, as it reflects the system's ability to correctly identify fraudulent transactions. While the autoencoder maintains a fixed recall of 80% across different thresholds, the hybrid system achieves a higher recall of 92%. As shown in Table III, this comes with some trade-offs in terms of precision and accuracy. Some sensitive rules (e.g., detecting unusual transaction hours or rare merchant categories) can flag borderline cases as fraud, but may still be legitimate. Besides, when there is a high-risk clue, LLM often leans toward flagging the transaction as risky instead of rejecting it, especially when it matches certain rules. In real-world fraud detection, prioritizing recall over precision is acceptable because missing fraud is costlier than reviewing a false positive [12].

TABLE III
PERFORMANCE COMPARISON BETWEEN AUTOENCODER AND RAG-BASED HYBRID APPROACH

Metric	Autoencoder			Hybrid System	
	85th	90th	95th	gpt-4o	gemini-2.5 -pro
Accuracy	82.00%	84.00%	86.00%	60.00%	76.00%
Precision	83.33%	86.96%	90.91%	56.10%	69.70%
Recall	**80.00%**	**80.00%**	**80.00%**	**92.00%**	**92.00%**
F1 Score	81.63%	83.33%	85.11%	69.70%	79.31%

2) Confusion Matrix Analysis: According to the confusion matrices, the autoencoder identifies 20 fraudulent transactions while missing 5, whereas the RAG-based hybrid approach identifies 23, demonstrating improved detection. As shown in Fig. 3, the RAG Gemini model (AUC 0.83) outperforms RAG GPT (0.65), and although baseline models reach 0.94, Gemini 2.5 surpasses GPT-4o due to stronger contextual understanding and superior structured reasoning for complex fraud detection [19]. While AE-RAGX shows lower overall accuracy than the standalone autoencoder, accuracy can be misleading in fraud detection due to class imbalance; the recall improvement from 80% to 92% is more critical, as even small gains can prevent thousands of undetected frauds in large-scale deployments. Moreover, AE-RAGX provides explainable outputs, enabling analysts to validate and act on alerts more effectively.

3) Explanation: In addition to the recall improvement, the proposed RAG-based hybrid system offers interpretation by providing rule-based explanations for each flagged transaction. Unlike the autoencoder, which operates as a black-box model, the RAG architecture retrieves relevant rules or patterns from the knowledge base. These rules justify its predictions, making the decision process transparent and

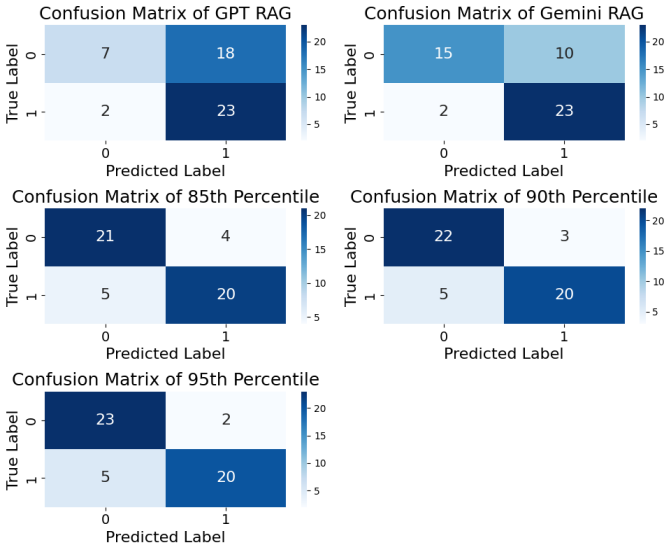

Fig. 2. Confusion Matrix for the Different Approaches

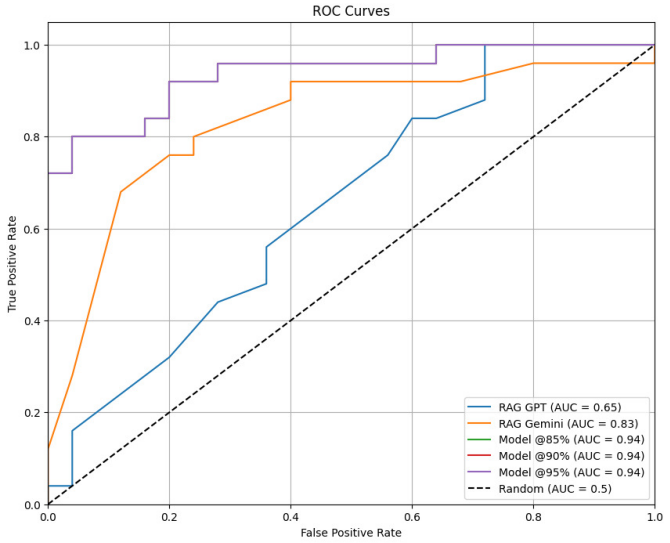

Fig. 3. ROC Curves for the Different Approaches

understandable to end-users. This explainability is possible because the LLM can combine statistical anomaly signals with human-curated domain knowledge, allowing it to articulate the specific behavioral or contextual violations that triggered the fraud flag. Such clarity not only helps analysts verify decisions more efficiently but also builds trust in the system, which is critical in high-stakes applications like fraud detection. Some generated reasons from the system for both fraud and clean transactions are shown in Table IV.

Our implementation averaged 1025 input tokens and 100 output tokens per request, with cost depending on the LLM provider's pricing model. Unlike conventional models, AE-RAGX incurs token costs but offers explainability and improved recall, which are critical in fraud detection (cf. [16]). As shown in Table V, AE-RAGX introduces slightly higher latency (2–3 seconds) due to autoencoder inference combined with retrieval and generation. However, this remains acceptable within typical fraud detection pipelines, where several seconds are available for verification. Crucially, AE-RAGX improves recall from 80% to 92% and provides rule-based reasoning interpretable by analysts. This trade-off is valuable: the prevention of false negatives outweighs the marginal processing delay, given the financial and reputational risks of undetected fraud.

TABLE V
AE-RAGX COMPARISON WITH STATE OF THE ART MODELS

Feature	AE-RAGX	OC-SVM [16]	Random Forest [17]
Accuracy	Moderate (60-76%)	High (70%)	High (99.5%)
Latency	Higher (2-3s)	Low (0.01-0.12s)	Low inference time
Complexity	High	Moderate to High: depends on kernel and data size	High: depends on number of trees
Explainability	Very High: human-readable explanation	Low: black-box	Moderate Explainability

VI. CONCLUSION

This research introduces AE-RAGX, a novel hybrid anomaly detection framework that addresses three critical limitations of traditional methods: lack of explainability, inability to capture contextual information and long-range dependencies, and suboptimal recall performance in critical applications. By integrating traditional autoencoders with RAG-based Large Language Models, our approach restores the missing contextual awareness that statistical methods cannot capture while providing transparent, rule-based explanations for each anomaly detection decision.

The experimental results on the credit card fraud detection dataset show that the hybrid model improves recall compared to a standalone autoencoder. Although the hybrid system's accuracy is slightly lower, the increased recall is valuable in fraud detection, where minimizing false negatives is critical. The security is immediately improved by even small recall enhancements, which result in large-scale decreases in undetected fraud. By combining statistical anomaly scores with domain-specific rules and contextual reasoning, the system can detect anomalies that traditional models might miss, while also enhancing trust through explainable outputs.

Our hybrid approach successfully improves recall, which is essential in fraud detection scenarios where reducing false negatives directly impacts financial security. While the trade-off results in a slight reduction in overall accuracy, this is acceptable and preferable in high-stakes anomaly detection settings where missing anomalies has greater cost than additional verification. Additionally, the system successfully achieves explainability by providing rule-based justifications for each detection, addressing the critical transparency gap in traditional anomaly detection methods. Future work will focus on optimizing the RAG-LLM component for higher accuracy and extending the approach to other domains.

TABLE IV
REASONS GIVEN FROM THE SYSTEM FOR 4 DIFFERENT TRANSACTIONS

Prediction	Reason
Fraud	The transaction was flagged as suspicious due to several factors: unknown location activity with a travel distance from home of 83.96 miles, making it a high-risk transaction under Location-Based Rules. Additionally, it features an unknown merchant location due to coordinates vs location consistency check needed. There was also an anomalous reconstruction error of 2.881648, which is significantly higher than the threshold of 0.35. Combined with the distance from the cardholder's usual location and the late transaction hour, these factors increase the likelihood of fraud.
Fraud	The transaction matches multiple high-risk criteria: the amount is significantly higher than the average, it occurred at an unusual hour (midnight), and the location is an unknown area 44.47 miles away from the cardholder's home. Additionally, the reconstruction error is 0.258825, indicating anomaly detection concerns, though the score alone does not meet the fraud threshold. Analysis of the behavior pattern with a critical score accumulation due to multiple high rules and unusual spending behavior leads to a fraud determination.
Clean	No critical or high fraudulence rules met. Although the transaction amount is higher than the 30-day average, the increase isn't drastically high. No pattern of high-frequency transaction activity and the anomaly reconstruction error is low (0.081178). Coordinates consistency needs verification, but no immediate fraud is detected based on available details. 0.081178
Clean	No critical or high-risk rules matched; a medium risk rule was identified for unknown location activity but not enough to warrant fraud determination. The reconstruction error is below critical threshold (0.273926 ¡ 0.45).

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